

One Capability or Many?

Testing the Economic Validity of Frontier AI Evaluation

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Abstract

Frontier-model leaderboards increasingly report “economic” benchmarks that aim to measure performance on professionally valuable work, such as software engineering, banking workflows, and other job-like tasks. These benchmarks may capture a distinct economic capability, or they may re-express the general capability signal that all benchmarks share as models improve. The distinction is a question of construct validity, and no prior work has tested it directly. We address it on a pinned snapshot of 421 frontier model configurations, of which 96 carry the complete twelve-benchmark battery the structural analysis requires. We frame benchmark validity as a latent-variable measurement problem, and we combine exploratory factor analysis under an oblique rotation with a leave-one-benchmark-out predictive test, assessing structural and predictive evidence against pre-registered thresholds. A single factor accounts for 74.5% of common variance and is strongly associated with model release date ($R^2 = 0.505$). Residualising on release date reduces that share to 59.6%, a 14.9-point drop. Its bootstrap interval straddles our 15-point discriminance threshold, so the effect is indistinguishable from the threshold at this sample size. Our dimensionality rule retains a single factor, and we report the registered distinctiveness test as not supported. When we extract more factors than that rule selects, however, the economic benchmarks do separate, and the separation carries out-of-sample weight, improving prediction of held-out economic scores over a single general-capability index on three of the four economic benchmarks (pooled $\Delta\text{MSE} = 0.037$, 95% bootstrap interval $[0.019, 0.055]$). Economic benchmarks therefore carry limited, incremental validity. They refine a dominant progress factor at the margin and remain largely explained by it. All hypotheses are pre-registered, the failing and near-threshold tests are reported as such, and every input is hash-pinned for exact reproduction.

Keywords: *Large Language Models; Benchmarking; Construct validity; Factor analysis; Predictive Validity; Evaluation*

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1 Introduction

1.1 Motivation and research questions

Public leaderboards have become the dominant instrument for comparing, procuring, and governing frontier language models. A recent shift in what they report now raises a measurement question with direct economic stakes. Alongside academic knowledge tests such as GPQA and Humanity’s Last Exam, leaderboards report *economic* benchmarks such as GDPval, the τ -Bench family, and terminal or agentic coding suites, all built to proxy paid professional work and multi-step tool use. Whether these economic benchmarks deserve a separate column depends on what they measure. A capability distinct from general test-taking would carry information that a single “intelligence” score cannot, and separate reporting would be warranted. A column that merely re-expresses one dominant axis of model progress would overstate what the leaderboard knows. This question concerns construct validity, the degree to which a measurement captures the construct it claims to. Recent work finds construct validity systematically under-examined in AI evaluation [1, 2].

We make the question empirical and answer it with two machine-learning tasks under a pre-registered design. We frame a snapshot of frontier models scored across twelve benchmarks as a psychometric problem, in which benchmarks are items, models are respondents, and scores are the observed responses from which latent capability structure is inferred. Two research questions organise the analysis. The first concerns structure. How many latent capabilities does the battery measure? Is a dominant axis an artefact of release date and scale, and do the economic benchmarks hold a separable position once that axis is controlled? The second concerns prediction. Does a multi-factor representation forecast a model’s held-out economic-benchmark score more accurately than a single general-ability index? The first question asks whether an economic factor exists in the correlational structure. The second asks whether any such factor carries information that survives out of sample.

1.2 Related work

Our study joins a small but growing literature that treats LLM benchmarks as measurement instruments open to measurement error. Bean et al. [1] review 445 benchmarks and find that only a minority justify their construct validity or apply any statistical test. Their finding motivates the psychometric scrutiny we attempt here. Kearns [2] makes that scrutiny quantitative. Fitting factor models to the Open LLM Leaderboard, Kearns shows that a dominant latent factor explains roughly 72% of communal variance and correlates with model scale, and that partialling scale out redistributes the variance across several factors. That 72% $\rightarrow$ 41% correction supplies both the template for our residualisation and an external calibration point for our own numbers. Kearns studies academic suites, however, and does not isolate the economic column we target. Our residualisation corrects the same uncontrolled single-factor reading, named in the registered plan as robustness item R4.

A second strand concerns how benchmark performance drifts over time. Akhtar, Reuel et al. [3] find that benchmark age and test-set scale are the strongest predictors of lost discriminative power. Release date is therefore a first-order confound when capabilities are compared across a moving frontier, and we residualise on it before interpreting any factor structure. Reuel, Ghosh et al. [4] map coverage gaps across social-impact evaluations and argue that the dimensions a suite surfaces are themselves a validity question. Their framing supports our focus on whether economic capability is measured separately or folded into a general score.

Methodologically, the contribution is the measurement design, built from established multivariate tools. These are exploratory factor analysis with oblique rotation [7], parallel analysis [5] and the scree criterion [6] for dimensionality, sampling-adequacy diagnostics [8, 9], silhouette [10] and gap-statistic [11] cluster validation, regularised and tree-based regression [12, 13, 14, 15], SHAP attribution [16], and the bootstrap [17]. The regression learners and cross-validation use scikit-learn [19]. The factor analysis uses the `factor_analyzer` package, and SHAP its own reference implementation [16].

1.3 Contributions

The paper makes four contributions, each addressing a gap the literature leaves open. First, it asks whether the *economic* column of a frontier leaderboard measures a capability distinct from general test-taking. Earlier factor-analytic studies [2] examined academic suites and treated capability as a single undifferentiated

construct. The economic benchmarks that now drive procurement and policy claims have never been subjected to a construct-validity test, and that target is the paper’s central novelty. Second, the paper isolates model release date as a first-order confound and residualises on it before reading any factor structure. What looks like one dominant capability axis then proves substantially a temporal artefact. Third, the paper introduces a leave-one-benchmark-out predictive-validity test that recasts the distinctiveness question as an out-of-sample forecast. This criterion is stronger and more decision-relevant than in-sample loading patterns, because a benchmark that adds nothing to prediction of held-out economic performance adds nothing a practitioner can use. Fourth, the paper delivers the whole analysis under a pre-registered protocol with hash-pinned inputs. It reports the near-threshold H2(ii) result and the not-supported H3 verdict alongside the confirmed ones, and documents every deviation from the registered plan.

1.4 Scope and non-goals

Fixing scope in advance keeps the claims disciplined and guards against over-reading a cross-sectional, observational result. In scope is a construct-validity analysis of the economic benchmarks on a single hash-pinned leaderboard snapshot. It uses exploratory factor analysis and clustering for structure and a leave-one-benchmark-out regression for predictive validity, under four pre-registered hypotheses with pre-specified thresholds. Several extensions fall outside it. The analysis is exploratory, so it claims no validated measurement model on fresh data, and it works from aggregate scores in place of item-level responses. It tests internal economic validity, meaning correlational and predictive structure, and leaves external economic impact such as professional usage or revenue to later work. It analyses one cross-section and models no saturation dynamics. Finally, it treats published scores as given and makes no causal claim. The residualisation identifies date as a strong correlate of the general factor without establishing cause.

2 Data

2.1 Source and provenance

The primary data set is a snapshot of the Artificial Analysis model leaderboard [20], captured on 6 July 2026 directly from the public leaderboard page and pinned by its SHA-256 hash (6f19f8...04de6; the full manifest appears in Appendix I). The snapshot contains 548 model configurations with per-benchmark scores, release dates, and provider metadata. A secondary source, the Epoch AI Notable AI Models data set (CC-BY 4.0) [21], supplies training-compute figures for a subset of models and serves only a scale robustness check. Both snapshots travel with the submission bundle, so every number in this paper is reproducible from fixed inputs, and neither source requires re-running a model.

2.2 The benchmark battery

We analyse the twelve benchmarks that survive the coverage rule of Section 2.3, grouped into five capability blocks. Four are economic benchmarks in the sense of proxying paid professional or agentic work, and the remainder span academic knowledge, scientific coding, long-context retrieval, and instruction following. Appendix D Table 4 describes each benchmark, the construct it targets, and its coverage in the analysis snapshot.

2.3 Inclusion rule and analysis grids

To avoid selecting benchmarks or models on the very outcomes under study, we fixed the coverage rule in advance. A benchmark is retained when at least sixty models carry a score, and a model is retained when it carries at least eight of the thirteen sufficiently-covered benchmarks. Thirteen of fourteen candidate benchmarks pass. We drop one agentic benchmark, APEX-Agents with twenty-six models, for sparsity, and demote one multimodal benchmark, MMMU-Pro, to a sensitivity-only role. Twelve primary benchmarks and 421 model configurations remain. These 421 rows are configurations, as a single base model can appear under several reasoning or effort settings. The registered analysis deduplicates to one row per base model as robustness check R1 (Appendix H), and we keep the configuration level for the primary analysis as the more conservative choice.

The economic benchmarks are scored on far fewer models than the academic core, so a single grid cannot serve both analyses and we work with two. The complete-case grid of 96 models scored on all twelve benchmarks is the only grid on which the economic block is complete, and it carries all structure hypotheses. The dense grid of 409 models scored on the nine near-universal benchmarks serves the model clustering, where sample size matters more than economic completeness. This completeness-versus-sample trade-off is the central data-design decision, and we flag it wherever it bears on a result.

2.4 Preprocessing and encoding

Three preprocessing decisions shape every downstream result. The benchmarks are reported on incomensurable scales. Some are accuracy fractions, one is a hallucination-penalised score, and GDPval is an Elo rating.¹ We therefore z-standardise each benchmark before any analysis, using training-fold statistics only in the supervised task. Standardisation keeps the variance decomposition interpretable and stops the widest-range benchmark from dominating the covariance. We treat GDPval as a continuous target on its own standardised scale and exclude it from the accuracy-only logit-transform check. The rung (v) covariates enter as one-hot indicators for reasoning capability and open-weights status, log-parameters for parameter count, and a binary flag for undisclosed counts. Missingness is thereby modelled as information and never silently imputed.

3 Exploratory data analysis

We conduct exploratory analysis on the modelling data before fitting any model. The marking guidance calls for it, and the choices it surfaces, meaning which grid to use, whether to residualise, and how many factors to expect, should follow from the data. We examine coverage and missingness, correlation structure, and temporal patterns, summarised in Figure 1. All standardisation and model fitting in later sections uses the training portion of each split only, so the exploratory views here characterise the data without leaking test information into the models.

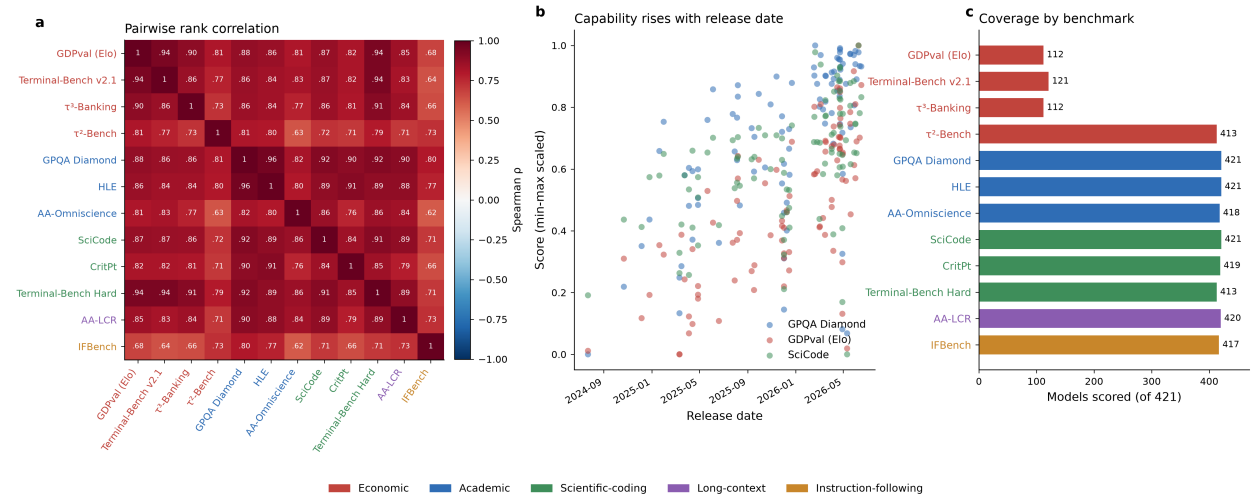


Figure 1: Exploratory data analysis of the twelve-benchmark battery on the complete-case grid ($n = 96$). (a) Pairwise Spearman rank correlations, with benchmark labels coloured by capability block, where almost every pair is strongly positive. (b) Per-benchmark scores (min–max scaled) against model release date, showing capability rising steeply with time. (c) Coverage per benchmark across the 421 retained models, with the four economic benchmarks (red) scored on far fewer models than the academic core.

Missingness is structural, as newer agentic benchmarks are run on fewer models (Figure 1c). We therefore use complete-case analysis on the two grids of Section 2.3 and impute nothing we would then be modelling. On the complete-case grid the Kaiser–Meyer–Olkin measure is 0.933, well above the 0.8 threshold [8], and Bartlett’s test of sphericity is decisively rejected [9]. Factor analysis is therefore appropriate.

¹GDPval Elo ratings come from pairwise human preference comparisons on economically valuable tasks, so a higher rating means more often preferred, on a scale with no fixed maximum.

The battery is highly collinear before any modelling. The mean off-diagonal Spearman correlation is $\rho = 0.79$, every benchmark pair is positive, and the economic benchmarks correlate with the academic core almost as strongly as with one another (Figure 1a). A dominant general factor would produce this near-block structure, and the structure analysis must explain it. A strong positive manifold is equally consistent with one underlying capability and with several capabilities that improve together over time. Distinguishing the two readings is the purpose of the residualisation in Section 4.1. The temporal view makes the ambiguity concrete. Scores rise steeply and jointly with release date (Figure 1b), so release date confounds any claim that two benchmarks measure the same thing. The saturation literature identifies model and benchmark age as the strongest predictor of capability change [3], and we therefore treat date as a first-class nuisance variable and residualise on it before interpreting factor structure.

Most benchmarks are bounded accuracies in $[0, 1]$, and several are strongly right-skewed, with CritPt at Fisher skewness 3.8, HLE at 1.5, and τ^3 -Banking at 0.9, their scores piling up near the floor (Appendix Figure 7). Two benchmarks are unbounded, GDPval on an Elo scale and AA-Omniscience on a penalised score, and both sit closer to symmetric. The bounded skewed accuracies motivate the logit-transform robustness check in Section 5.7, which confirms that the first-factor share is no artefact of the raw scale. The low-scoring tails hold weak models with genuinely low scores. We therefore retain every row and handle the boundedness through the transform, trimming no outliers. Therefore the univariate distributions justify the transforms used later.

4 Methods

The analysis runs two tasks on the same data. Task 1 is unsupervised and recovers and stress-tests the latent capability structure with three method families, namely principal-component analysis, exploratory factor analysis, and clustering. Task 2 is supervised and tests whether that structure has out-of-sample predictive value, using four regression learners under a leave-one-benchmark-out protocol. All features are per-benchmark z-standardised using training-set statistics only, and the full mathematical derivations are collected in Appendix C.

4.1 Task 1: latent structure

We estimate dimensionality with Horn’s parallel analysis [5], retaining factors whose eigenvalue exceeds the 95th percentile of eigenvalues from random data of the same shape. The scree criterion [6] serves as a visual cross-check. Parallel analysis is primary, as it is the more conservative of the two on strongly collinear data. We prefer factor analysis to a PCA-only description. PCA components mix common and unique variance, while the common-factor model separates shared capability from benchmark-specific idiosyncrasy. A construct-validity question turns on that distinction.

The factor model is a maximum-likelihood exploratory factor analysis with an oblique oblimin rotation [7]. Capability dimensions have no reason to be uncorrelated,² and the residualised factors are in fact strongly correlated (Section 5.3), so we choose the oblique rotation. The model represents each standardised benchmark score as a linear combination of k latent factors plus a benchmark-specific uniqueness, derived in Appendix C. We report the first factor’s share of common variance as the dominance statistic, and we orient every factor so that higher means more capable.

To separate a general capability from a shared time trend, we regress each standardised benchmark on release date, and on a subsample also on training compute, then re-fit the factor analysis on the residuals. The change in the first factor’s variance share measures how much of the apparent general factor is a date artefact, directly comparable to the 72% $\rightarrow$ 41% correction Kearns reports [2]. This residualise-then-refit design lets the analysis separate a single date-driven manifold from a truly multidimensional capability space.

We then cluster models in factor-score space with k -means, Gaussian mixtures, and agglomerative Ward linkage, selecting the cluster count by silhouette [10] and the gap statistic [11]. Three algorithms guard against a clustering that reflects one algorithm’s inductive bias. We cluster the benchmarks separately by correlation distance $1 - \rho$ under average linkage, testing whether the economic benchmarks form a distinct branch.

²An orthogonal rotation such as varimax forces the factors to be uncorrelated. On data with a strong general factor this spreads shared variance across several artificially independent axes and obscures the dominance we are trying to measure.

4.2 Task 2: predictive validity

We then recast the distinctiveness question as a forecast, asking whether a model’s score on a held-out economic benchmark follows from its behaviour on the others. The leave-one-benchmark-out (LOBO) protocol takes each economic benchmark as the target in turn and predicts it from representations of the remaining eleven on the $n = 96$ complete-case grid. Success therefore demands transfer across benchmarks. A conventional random split would let a learner memorise a single column, and the LOBO design forecloses that. A five-rung predictor ladder of increasing richness makes the comparison concrete. Rung (i) uses release timing and scale only, rung (ii) a single mean-score general index, rung (iii) the first factor only, rung (iv) the k factors, and rung (v) the k factors plus model covariates. The contrast of interest sets rung (iv) or (v) against the single-index rung (ii). A multi-factor representation that beats a single general index out of sample would show that economic capability carries information the index does not.

We fit each rung with four learners spanning the linear-regularised and non-linear tree families [18], namely ridge [12], elastic net [13], random forest [14], and gradient boosting [15], so that the ladder comparison is not tied to a single model class. We tune hyperparameters by grid search in an inner five-fold cross-validation, nested inside an outer five-fold split that supplies the reported out-of-fold error. No test fold then informs tuning or standardisation. A single non-nested cross-validation would let tuning peek at the evaluation folds and inflate every rung. We report both train and test RMSE and out-of-fold R^2 .

The factors are estimated quantities, so we re-estimate them inside the loop. We fit the factor analysis on each training fold alone, project the held-out fold through the training-fold Thurstone factor-score weights $W = R^{-1}\Lambda$ (Appendix C), and re-orient it by the training-fold mean index. The factor construction therefore never sees the models it is evaluated on, and a naive fit-on-all-data pipeline would silently open that leakage path.

For each economic target we compute $\Delta\text{MSE} = \text{MSE}(\text{single index}) - \text{MSE}(k \text{ factors})$, so that positive values favour the multi-factor representation, and we obtain a 95% confidence interval by bootstrap resampling the pooled out-of-fold errors [17] with $B = 2000$ replicates. H4 holds when the pooled interval excludes zero and a majority of the four economic targets individually favour the multi-factor model. Figure 4 in Appendix A states the complete procedure, with the leakage-safe factor re-estimation inside every fold.

4.3 Hypotheses and pre-specified tests

The design fixes four hypotheses with quantitative thresholds before analysis, calibrated to the 72% $\rightarrow$ 41% result of Kearns [2]. H1, dominance, holds that the first factor carries more than 50% of common variance. H2, the date artefact, has two parts. The first is that the dominant factor’s release-date R^2 is at least 0.30 and the strongest of the factors. The second is that residualising on date lowers the first factor’s share by at least fifteen percentage points. H3, economic distinctiveness, holds that after residualisation the four economic benchmarks load at least 0.40 on a common factor and exceed their cross-loadings. H4, predictive validity, holds that the multi-factor representation’s ΔMSE confidence interval excludes zero. We report every outcome. This includes H2(ii), which becomes indistinguishable from its threshold once uncertainty is attached, and H3, whose registered retention clause is not met.

Appendix B sets out the study’s positionality and directionality, covering factor sign orientation, the direction of residualisation, the pre-fixed sign of the predictive contrast, and the two sampling threats to validity.

4.4 Evaluation metrics and model selection

In model evaluation, it is important to match metrics to each task’s aim. For the unsupervised structure task the primary quantities are the first-factor communal-variance share for dominance, each factor’s release-date fit for temporal confounding, and the residualised loading pattern for distinctiveness. We check model adequacy beforehand with the Kaiser–Meyer–Olkin index and Bartlett’s test. We choose the number of factors by parallel analysis, as information criteria over-retain on strongly collinear data. Two counts appear in the paper. The primary count is $k = 1$, returned by parallel analysis, the Kaiser rule, and the scree elbow. The Bayesian information criterion alone prefers $k = 4$, which we treat as over-retention. A three-factor solution probes economic distinctiveness, and no criterion selects it. It is an exploratory over-extraction, and it asks whether an economic block would separate once more factors than the data support are forced out. The choice of

three is a documented deviation from the pre-registered plan (Appendix G). The predictive conclusion does not hinge on it. Sweeping the k -factor predictor over $k \in \{2, 3, 4, 5\}$ beats the single index at every count, with all bootstrap intervals excluding zero and the gain growing with k (Appendix H). At rung (iv), Task 2’s k -factor predictor denotes this same three-factor solution.

For the supervised task the objective is mean squared error on standardised targets. We report root mean squared error, which stays on the target’s own scale and remains comparable across rungs and targets. Threshold metrics such as ROC and its area do not apply, as neither task is a binary classification. Model selection is strictly nested. The inner cross-validation chooses hyperparameters and the best learner per rung, and the untouched outer cross-validation supplies the reported error. We report both train and test error in Table 2, which exposes any over-fitting of the kind the timing-only rung shows.

5 Results

In sum, the four hypotheses are reported in order. Sections 5.1 to 5.3 address the structure question. They cover the dominance of a single factor (H1), the extent to which it is a release-date artefact (H2), and whether the economic benchmarks form a distinct factor once that artefact is removed (H3). Section 5.4 corroborates the structure with clustering, Section 5.5 turns to prediction (H4), Section 5.6 opens the best model with explainability and named error analysis, and Section 5.7 collects the robustness checks. Each hypothesis is adjudicated against its pre-registered threshold, and the outcome is reported regardless of direction.

5.1 Dominance of a single factor (H1)

Every dimensionality diagnostic points to a dominant capability axis. The first principal component alone accounts for 79.4% of total score variance, and parallel analysis retains a single factor. Only the first observed eigenvalue of 9.53 exceeds its random-data 95th percentile of 1.79, and the second, at 0.91, already falls below (Appendix Figure 9). In the maximum-likelihood factor analysis the first factor holds 74.5% of common variance, above the 50% dominance threshold and close to the 72% Kearns reports on a different leaderboard [2]. H1 is therefore supported. The share is robust to the correlation measure. A rank-based factor analysis gives 94.5% and a logit-transformed one 91.8%, so the Pearson-based headline is the conservative choice (Section 5.7).

5.2 A date-driven artefact (H2)

Having established a dominant factor, we ask whether it reflects stable capability or the passage of time. The factor’s scores track model release date with a logistic-fit R^2 of 0.505, the strongest of the three factors against 0.364 and 0.286. H2(i) is therefore supported, and the leading axis of “capability” is substantially a time trend. Residualising every benchmark on release date and re-fitting then lowers the first factor’s common-variance share from 74.5% to 59.6% (Figure 2c), a drop of 14.9 percentage points, so date alone accounts for about a fifth of the general factor. The point estimate falls just short of the pre-registered fifteen-point threshold. A bootstrap over models puts a 95% interval of $[-5.3, +32.7]$ around it. On the primary grid the drop is therefore indistinguishable both from zero and from the threshold, and we report H2(ii) as not met on the point estimate, without moving the line after seeing the result.

The verdict depends on the unit of analysis, and the main text should say so. Under the registered base-model unit, which keeps one row per model, the drop is 24.1 points and clears the threshold. Under the deviated configuration-level primary it is 14.9 points and fails. On the 58-model compute-known subsample, date alone gives 16.5 points and also clears it. Two of the three specifications pass. We keep the failing configuration-level specification as primary, as it is the more conservative reading. The confirmation of a temporal confound then rests on the subsample and the deduplicated grid, and the primary grid alone does not carry it. The residual 59.6% share is what any downstream capability claim must contend with.

Adding training compute to the residualisation does not deepen the correction. On the compute-known subsample the date-only drop is 16.5 points, and the joint date-plus-compute drop is smaller at 9.3 points. A shrinking correction is an anomaly, and the likely mechanism is collinearity between release date and compute on this small subsample. Later models are also larger, so regressing on both predictors splits their shared

variance and leaves a smaller residual movement. Compute therefore adds little beyond date, and the date effect stands.

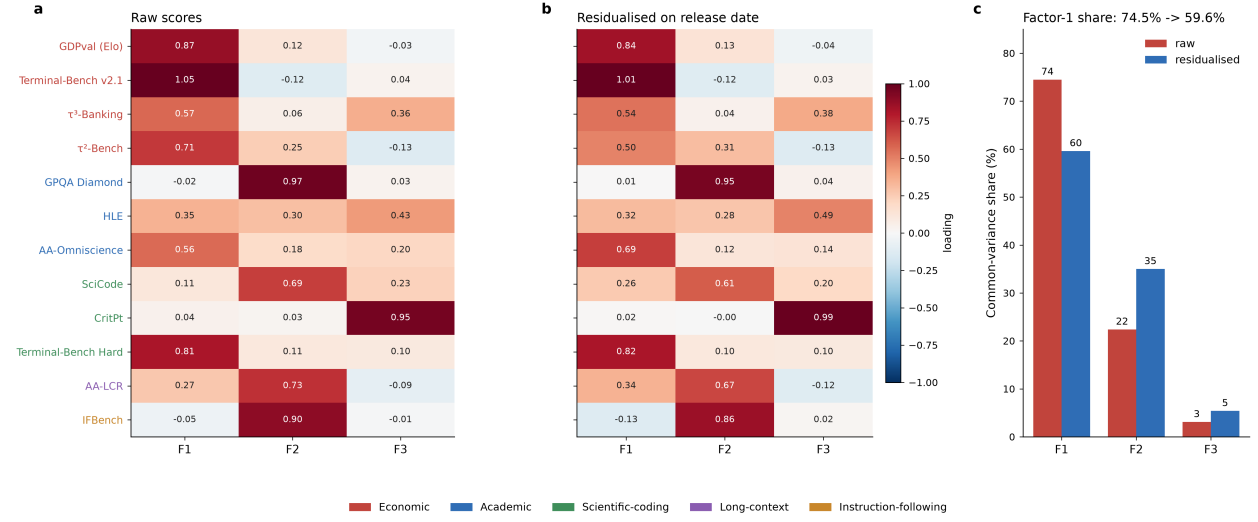


Figure 2: Factor loadings from the three-factor oblimin exploratory factor analysis. (a) On raw scores the economic benchmarks (bold) already concentrate on F1 under the three-factor rotation. (b) After residualising on release date, the economic benchmarks concentrate on their own factor F1, with a largest economic cross-loading of 0.38. (c) The first factor’s common-variance share falls from 74.5% to 59.6% under residualisation, a 14.9-point drop.

5.3 Economic distinctiveness (H3)

Having removed the temporal confound, we test whether the economic benchmarks form a distinct factor. The registered test for H3 conjoins two conditions. Parallel analysis must retain a factor, and that factor’s loadings must concentrate discriminantly on the economic block. The two conditions are not jointly met, and we report H3 as not supported as registered. Parallel analysis on the residualised data retains a single factor, and that one-factor solution leaves no block to separate. The discriminant separation appears only once three factors are extracted, and the dimensionality rule does not endorse that extraction. Holding H3 to its registration as H2(ii) was held to its 15-point line, the retention clause fails and the pre-registered negative-result clause applies.

Under a three-factor over-extraction the economic block does separate, and we report the pattern as exploratory. All four economic benchmarks load at least 0.40 on a single factor, F1, with GDPval at 0.84, Terminal-Bench v2.1 at 1.01, τ^3 -Banking at 0.54, and τ^2 -Bench at 0.50, against a largest economic cross-loading of 0.38 (Table 1). The mean absolute F1 loading is 0.72 for the economic benchmarks and 0.32 for the rest. The three factors are strongly correlated, at 0.67 between F1 and F3 and 0.72 between F2 and F3. The oblique rotation is therefore appropriate, and the separation reads best as a secondary refinement of one dominant axis, well short of independent dimensions. Bootstrap intervals on the economic loadings are wide, spanning 0.10 to 0.78 for τ^2 -Bench and 0.44 to 0.99 for GDPval, so the exploratory pattern is suggestive and loosely estimated. Out-of-sample transfer, however, rescues the economic factor from the charge of being an over-extraction artefact. As Section 5.5 shows, the three-factor representation predicts held-out economic scores better than a single index. Out-of-sample transfer is a stronger dimensionality argument than an in-sample retention rule, and we let the predictive test carry the claim the structural test cannot.

5.4 Clustering of models and benchmarks

A factor model can impose structure the raw data do not support, so we sought an independent check, and clustering corroborates the low dimensionality without assuming a factor model at all. Grouping models in factor-score space, all three algorithms prefer $k = 2$ by silhouette, at 0.480 for k -means, 0.499 for agglomerative, and 0.467 for the Gaussian mixture (Appendix Figure 5b). The gap statistic increases monotonically in k and finds no interior optimum, so it is inconclusive on data with a strong general factor and we set it aside. The two clusters mark capability tiers along one axis of overall strength. A larger, earlier, lower-scoring group of 302 models, with median release 2025-09 and mean Intelligence Index 13.7, separates

Table 1: Residualised (date-partialled) three-factor oblimin loadings, economic benchmarks in bold. F1 is an agentic and work-realistic factor, F2 an academic-knowledge factor, F3 a physics and hard-reasoning factor. All four economic benchmarks load highest on F1 with small cross-loadings.

Benchmark	F1	F2	F3
GDPval (Elo)	0.84	0.13	-0.04
Terminal-Bench v2.1	1.01	-0.12	0.03
τ^3 -Banking	0.54	0.04	0.38
τ^2 -Bench	0.50	0.31	-0.13
GPQA Diamond	0.01	0.95	0.04
HLE	0.32	0.28	0.49
AA-Omniscience	0.69	0.12	0.14
SciCode	0.26	0.61	0.20
CritPt	0.02	-0.00	0.99
Terminal-Bench Hard	0.82	0.10	0.10
AA-LCR	0.34	0.67	-0.12
IFBench	-0.13	0.86	0.02

from a smaller, newer, higher-scoring group of 107 models, with median release 2026-03 and mean index 37.1. The split sorts models by how much capability they hold, along the same date-driven axis (Appendix Figure 5a). Clustering the benchmarks by correlation distance places ten of twelve, and three of four economic benchmarks, in one block, with τ^2 -Bench and IFBench branching separately (Appendix Figure 10).

5.5 Predictive validity (H4)

Table 2: Predictor-ladder cross-validated errors (economic block). Train and test RMSE are on standardised targets and test R^2 is out-of-fold. The best learner per rung is shown. The k -factor rungs (iv, v) achieve the lowest test error. Errors are pooled over the five outer folds; the uncertainty on the key rung (ii)-versus-(iv) contrast is the bootstrap Δ MSE interval in Table 3, which excludes zero.

Rung	Best learner	Train RMSE	Test RMSE	Test R^2
(i) timing / scale	random forest	0.521	0.741	0.459
(ii) mean index	ridge	0.463	0.474	0.771
(iii) first factor	elastic net	0.933	0.950	0.110
(iv) k factors	ridge	0.410	0.433	0.808
(v) k factors + covariates	ridge	0.391	0.438	0.800

Structural separation alone does not make the economic factor useful, so we ask whether it predicts held-out economic performance better than a general index. Across the predictor ladder a single mean-score index is already a strong baseline, at economic-block test RMSE 0.474 and R^2 0.771. The k -factor representation improves on it to RMSE 0.433 and R^2 0.808, and adding covariates gives no further gain (Table 2, Figure 3a). The first factor alone predicts poorly at RMSE 0.950, which locates the economic signal in the secondary factors, where the over-extraction placed it. Bootstrapping Δ MSE over the pooled out-of-fold errors gives a pooled improvement of +0.037, with a 95% interval of [+0.019, +0.055]. That resample includes near-duplicate settings of the same base model, so we re-run it on the deduplicated grid of 89 distinct models, where the improvement is +0.038 with [+0.020, +0.056] (Appendix H). This model-level bootstrap carries the inference. Three of the four economic benchmarks improve individually, GDPval by +0.026 with [+0.010, +0.044], Terminal-Bench v2.1 by +0.028 with [+0.001, +0.056], and τ^3 -Banking by +0.056 with [+0.008, +0.103],

Table 3: H4 bootstrap Δ MSE = MSE(mean index) – MSE(k factors), $B = 2000$; positive favours the multi-factor representation. Intervals that exclude zero are marked $\checkmark$.

Target	Δ MSE	95% CI	Excl. 0
GDPval (Elo)	+0.026	[+0.010, +0.044]	$\checkmark$
Terminal-Bench v2.1	+0.028	[+0.001, +0.056]	$\checkmark$
τ^3 -Banking	+0.056	[+0.008, +0.103]	$\checkmark$
τ^2 -Bench	+0.038	[-0.011, +0.086]	
Pooled economic	+0.037	[+0.019, +0.055]	$\checkmark$

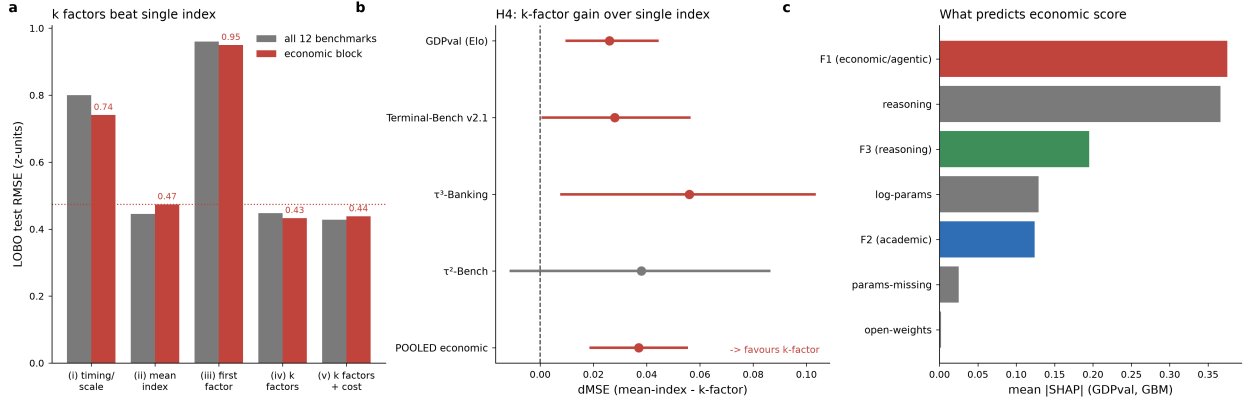


Figure 3: Predictive validity under the LOBO protocol. (a) Test RMSE by predictor-ladder rung, for the economic block (red) and all twelve targets (grey). The k -factor rungs beat both the single index and the first-factor-only rung. (b) Bootstrap Δ MSE (single index minus k factors) with 95% intervals, where positive favours the multi-factor representation. (c) SHAP importances for the GDPval model, showing the agentic factor F1 and the reasoning flag dominating.

while τ^2 -Bench’s interval includes zero (Table 3, Figure 3b). We report the per-target mean of +0.037 as a descriptive spread, as a percentile interval over four positive means cannot include zero by construction. H4 is therefore supported, on the pre-registered rule of a pooled interval that excludes zero with a majority of targets favouring the multi-factor model.

The effect is nonetheless modest. An improvement of 0.037 sits on top of a single index that already explains 77% of economic-block variance, so the multi-factor gain is incremental and leaves the model ranking largely intact. Its value lies in direction and consistency. The gain survives leave-one-benchmark-out validation, which rewards transfer across constructs and penalises memorisation, and it holds on three of four economic targets. The one exception, τ^2 -Bench, carries the widest interval, and its null reflects limited coverage as much as an absence of signal.

5.6 Explainability and error analysis

To interpret the predictive gain, we open the best model on the largest economic target. GDPval is predicted by a k -factor ridge with out-of-fold R^2 0.88. Permutation importance and SHAP attribution [16] agree that the agentic factor F1 is the dominant predictor, followed by the reasoning flag and the hard-reasoning factor F3, with scale and open-weights status contributing little (Figure 3c). We found a clean diagnosis. The predicted-versus-observed plot lies on the identity line, and the residuals show no systematic trend (Appendix Figure 8a,b). The largest residuals are interpretable outliers. They include a non-reasoning variant of a frontier model that outperforms its economic prediction, Grok 4.3 Non-reasoning at +1.14, and two models that under-perform theirs (Appendix Figure 8c). The pattern fits GDPval rewarding agentic behaviour that the other benchmarks capture only indirectly.

5.7 Robustness

The headline structure is stable under design perturbations. Full tables and a deviations-from-design account appear in Appendices G and H. The most consequential check is the registered configuration-versus-deduplication comparison. Collapsing to one row per base model guards against near-duplicate configurations inflating correlations, and it raises the first-factor share to 90.2% and the date-residualisation drop to 24.1 points. The configuration-level analysis is therefore the conservative choice. Replacing Pearson with rank-based or logit-transformed correlations raises the first-factor share to 94.5% and 91.8%, so H1 is no artefact of the correlation metric. On the compute-known subsample the date-only residualisation drops the share by 16.5 points, and adding log-compute gives a smaller drop. The predictor ladder’s ordering holds on both the economic block and all twelve targets, with the single index strong, the first factor alone weak, and the k factors best (Table 2). The selected hyperparameters are pluralities across targets (Table 6), and the H4 verdict does not depend on the exact regularisation strength.

6 Discussion and conclusion

The two tasks answer our proposed research question. The frontier benchmark battery is governed by one dominant capability axis, a first factor holding 74.5% of common variance. That axis is substantially a release-date artefact, however. Models improve on everything at once as they get newer, and partialling date out removes about a fifth of the general factor. In the raw data the economic benchmarks form no separate kind of capability. They sit inside the general cloud and only under an over-extraction that a strict dimensionality rule would not endorse. Even so, the secondary economic factor carries out-of-sample information, and it predicts held-out economic scores better than a single general index on three of four benchmarks. The registered negative-result clause anticipated this outcome. Economic validity is therefore limited. A date-driven progress factor accounts for most of the economic signal, and the benchmarks add a modest increment beyond it.

The reading matters for how leaderboards should be used. A single intelligence index captures most of what distinguishes models and gives a defensible first-order summary, yet the residual economic signal is the part most relevant to claims about real-world professional value. Reporting economic benchmarks separately is therefore justified, though small gaps between contemporaneous frontier models call for caution, as most of the apparent spread is the shared time trend. These findings reinforce the call for psychometric scrutiny of AI benchmarks [1, 2] and the warning that temporal effects are a first-order confound [3].

The result also points to how new economic benchmarks should be validated. When most of the spread on a benchmark is the shared date trend, the benchmark’s value for distinguishing contemporaneous models depends on the variance it adds once that trend is removed. A builder who wants to show that a new economic suite measures something distinct can apply the two tests here directly. Among models released in the same window, the builder can check that the benchmark’s residualised loading separates from the general factor and that it improves out-of-sample prediction of held-out economic tasks. This adds a predictive lens to the review-based [1] and saturation-based [3] approaches.

6.1 Limitations

Four limitations bound the claims. First, the design is exploratory, so it fits no confirmatory measurement model. The economic factor appears only under an over-extraction relative to parallel analysis, and a pre-registered confirmatory model on fresh data would test it more strictly. Second, the analysis is observational. The residualisation identifies date as a strong correlate of the general factor, yet it does not establish that date causes the shared variance, and unobserved common causes such as shared training data across contemporaneous models could produce the same pattern. Third, coverage is uneven. The economic benchmarks are scored on far fewer models than the academic core, so the complete-case grid of 96 models is small and the economic results carry wider uncertainty than the dense-grid clustering. Fourth, the results describe one leaderboard on one date and will shift as new models arrive, a risk we mitigate with a hash-pinned, fully reproducible snapshot.

In light of these limitations, we propose three directions of future work. First, a confirmatory factor analysis on a fresh snapshot, testing the over-extracted economic factor against a pre-specified model. A second links benchmark scores to external adoption measures, such as task-completion rates in deployment, moving the argument from internal structure to external impact. A third recovers the per-task token and turn counts the snapshot omits, letting rung (v) model cost and effort directly and sharpening the test of whether economic capability is separable from the resources a model spends.

6.2 Conclusion

Treating a frontier-model leaderboard as a psychometric measurement problem, we asked whether economic benchmarks measure one capability or many, and the answer is neither extreme. A single, largely date-driven general factor dominates them, and they retain a small, real, out-of-sample-predictive economic component. Economic benchmarks earn their place as a modest correction to a general progress score, well short of an independent axis of ability. By pre-registering four tests, reporting the near-threshold and not-supported verdicts as plainly as the confirmed ones, documenting every deviation, and pinning every input, we make this a transparent and reproducible case study in the construct validity of AI evaluation.

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Appendix

This appendix collects the material that supports the main text without interrupting its argument. It states the prediction algorithm (A) and the study’s positionality (B), gives the mathematical derivations behind the factor and prediction models (C), the full coverage audit behind the inclusion rule (E), the tuning grid (F), the robustness battery (H), the reproducibility and provenance record (I), and the supplementary figures referenced from the results (J). Each subsection opens with a one-line statement of what it supports and where the main text relies on it.

A Leave-one-benchmark-out procedure

Supports Section 4.2, which describes the protocol in words.

Procedure Leave-one-benchmark-out predictive validity with nested cross-validation
Input: standardised score matrix Z (models $\times$ benchmarks), economic targets T , learners L , rungs R . for each target $t \in T$ do partition models into 5 outer folds for each outer fold (train, test) do fit standardisation and EFA on train only; form Thurstone weights $W = R^{-1}\Lambda$ project test models through W ; orient factors by train mean index for each rung $r \in R$ and learner $\ell \in L$ do tune ℓ by inner 5-fold grid search on train; predict test, store out-of-fold error select best learner per rung by pooled out-of-fold RMSE compute $\Delta\text{MSE} = \text{MSE}(\text{rung ii}) - \text{MSE}(\text{rung iv})$; bootstrap $B = 2000$ for 95% CI Output: per-target and pooled ΔMSE with confidence intervals (H4).

Figure 4: The leave-one-benchmark-out procedure with leakage-safe factor projection and nested tuning.

B Positionality, directionality, and threats to validity

Supports Section 4.3, which summarises these commitments in one sentence. A factor solution is identified only up to sign, so we state the direction of every factor explicitly. We orient each factor to correlate positively with the model’s mean standardised score. “Higher” then always means “more capable”, and every loading in Table 1 reads in the same direction. The choice is not cosmetic. An un-oriented solution could report the same economic factor with flipped signs and invite the opposite interpretation. The direction of the residualisation is equally deliberate. We regress scores on release date and analyse the residuals, which measures how much benchmark co-movement survives once the release date is known. We do not run the reverse regression, as we make no claim that the general factor causes the calendar. Finally, we fix the sign of the predictive contrast in advance. We define ΔMSE as the single-index error minus the k -factor error, so that a positive value favours the richer representation, and we pre-register that H4 requires this positive interval to exclude zero.

Two threats to validity follow from the study’s vantage point. First, we observe only the models that vendors chose to submit to a public leaderboard. The sample is therefore a convenience sample of frontier systems, and the claims generalise to models on this leaderboard alone. Second, the economic benchmarks are newer and sparser, so any economic-specific structure is estimated from fewer observations and is more sensitive to the particular models that carry those scores. We quantify that limitation with the wide bootstrap intervals in Table 3.

C Mathematical derivations

Supports Sections 4.1 and 4.2, which state the factor and prediction models in words and defer their formal statement to here.

The exploratory factor-analysis model. Let $\mathbf{z} \in \mathbb{R}^p$ be the vector of p per-benchmark z-standardised scores for a model. The common-factor model represents it as a linear combination of k latent factors $\mathbf{f} \in \mathbb{R}^k$ plus a benchmark-specific uniqueness $\boldsymbol{\varepsilon}$,

$$\mathbf{z} = \Lambda \mathbf{f} + \boldsymbol{\varepsilon},$$

where $\Lambda \in \mathbb{R}^{p \times k}$ is the loading matrix. Assuming $\text{Cov}(\mathbf{f}) = \Phi$, $\text{Cov}(\boldsymbol{\varepsilon}) = \Psi$ diagonal, and $\text{Cov}(\mathbf{f}, \boldsymbol{\varepsilon}) = \mathbf{0}$, the model implies the covariance structure

$$\Sigma = \Lambda \Phi \Lambda^\top + \Psi.$$

The parameters are estimated by maximum likelihood under multivariate normality. We compute the dominance statistic for H1 on the unrotated solution, where the factors are orthogonal ($\Phi = I$) and each factor’s contribution to common variance is its sum of squared loadings, $\sum_{j=1}^p \lambda_{jm}^2$, the m -th diagonal entry of $\Lambda^\top \Lambda$. The first factor’s share of common variance is therefore

$$s_1 = \frac{\sum_{j=1}^p \lambda_{j1}^2}{\sum_{m=1}^k \sum_{j=1}^p \lambda_{jm}^2} = \frac{(\Lambda^\top \Lambda)_{11}}{\text{tr}(\Lambda^\top \Lambda)}.$$

On the complete-case grid the three factors carry sums of squared loadings 7.79, 2.34, and 0.32, so $s_1 = 7.79/10.45 = 74.5\%$. This is a factor-level share, distinct from a single benchmark’s communality $\sum_m \lambda_{jm}^2$, a diagonal entry of $\Lambda \Lambda^\top$.

Oblique rotation and factor scores. The maximum-likelihood solution is identified only up to rotation, so we apply an oblique oblimin rotation, which minimises a quartimin-type criterion on the loadings while allowing $\Phi \neq I$. Factor scores are then recovered by the Thurstone regression method,

$$\mathbf{W} = R^{-1} \Lambda, \quad \hat{\mathbf{f}} = \mathbf{W}^\top \mathbf{z},$$

where R is the sample correlation matrix. In the prediction task $\mathbf{W}$ is estimated on the training fold only and applied to the held-out fold, and it is the leakage-safe projection referenced in Section 4.2.

Date and scale residualisation. To separate general capability from a shared time trend, each standardised benchmark z_j is regressed on model release date d (and, in a subsample, on log training compute),

$$z_j = \beta_{0j} + \beta_{1j} d + r_j,$$

and the factor model is re-fitted on the residuals r_j . The change in the first factor’s variance share between the raw and residualised fits, $74.5\% \rightarrow 59.6\%$, is the H2(ii) statistic.

Nested leave-one-benchmark-out estimator. For an economic target t , the out-of-fold error of rung r is

$$\text{MSE}_r = \frac{1}{n} \sum_{i=1}^n (z_{it} - \hat{z}_{it}^{(-\kappa(i))})^2,$$

where $\hat{z}_{it}^{(-\kappa(i))}$ is the prediction for model i from a learner trained on the outer folds excluding the fold $\kappa(i)$ containing i , with hyperparameters chosen by an inner five-fold grid search on those training folds. The predictive-validity statistic contrasts the single-index rung against the k -factor rung,

$$\Delta \text{MSE} = \text{MSE}_{\text{ii}} - \text{MSE}_{\text{iv}}.$$

Bootstrap confidence interval. A 95% interval for ΔMSE is obtained by resampling the paired out-of-fold squared-error vectors with replacement $B = 2000$ times and taking the 2.5th and 97.5th percentiles of the resampled differences. H4 is supported when this interval excludes zero.

D Benchmark battery

Supports Section 2.2, which names the battery and defers the full description here.

Table 4: The twelve-benchmark battery, grouped by capability block, with the construct each targets and its coverage in the analysis snapshot (models scored, of 421 retained). Higher is better for all; GDPval is an Elo rating, the others accuracy-type scores.

Benchmark	Construct measured	Coverage
<i>Economic</i>		
GDPval (Elo)	Paid professional task quality	112
Terminal-Bench v2.1	Terminal / agentic task completion	121
τ^3 -Banking	Multi-step banking workflows	112
τ^2 -Bench	Tool-use agentic tasks	413
<i>Academic</i>		
GPQA Diamond	Graduate-level science QA	421
HLE	Humanity’s Last Exam (frontier knowledge)	421
AA-Omniscience	Broad knowledge, hallucination-penalised	418
<i>Scientific coding</i>		
SciCode	Research-level scientific coding	421
CritPt	Physics problem solving	419
Terminal-Bench Hard	Hard terminal coding tasks	413
<i>Long-context / instruction</i>		
AA-LCR	Long-context reasoning	420
IFBench	Instruction following	417

E Full coverage audit

Supports the inclusion rule of Section 2.3. Table 5 reports every candidate benchmark on the raw snapshot and the decision applied to it.

Table 5: Full candidate-benchmark coverage on the raw 548-model snapshot, with the inclusion decision. Twelve benchmarks are primary, MMMU-Pro is demoted to sensitivity-only for sparse and multimodal coverage, and APEX-Agents is dropped below the sixty-model floor.

Benchmark	Block	Models scored	Role
GPQA Diamond	Academic	513	primary
HLE	Academic	509	primary
AA-Omniscience	Academic	418	primary
MMMU-Pro	Academic	204	sensitivity only
τ^2 -Bench	Economic	428	primary
Terminal-Bench v2.1	Economic	121	primary
GDPval (Elo)	Economic	117	primary
τ^3 -Banking	Economic	112	primary
APEX-Agents	Economic	26	dropped (sparse)
IFBench	Instruction-following	437	primary
AA-LCR	Long-context	441	primary
SciCode	Scientific-coding	507	primary
CritPt	Scientific-coding	422	primary
Terminal-Bench Hard	Scientific-coding	421	primary

F Hyperparameter grid and selections

Supports Section 4.2 and the robustness claim of Section 5.7 that the H4 verdict does not hinge on a single tuning choice. The inner grid searched ridge and elastic-net $\alpha \in \{0.03, 0.1, 0.3, 1, 3\}$ with elastic-net ℓ_1 ratio in $\{0.2, 0.5, 0.8\}$, random-forest depth in $\{4, 8, \text{none}\}$ with $\{200, 300\}$ trees, and gradient-boosting depth in $\{2, 3\}$ with learning rate in $\{0.05, 0.1\}$. Table 6 reports the modal selection per rung.

Table 6: Modal hyperparameters selected by the inner five-fold grid search, reported as the plurality choice across all twelve leave-one-benchmark-out targets with the agreement count. The H4 verdict rests on fold-averaged out-of-fold errors, not on any single setting.

Rung	Learner	Modal hyperparameters	Agreement
(i) timing / scale	random forest	max_depth=4, min_samples_leaf=3, n_estimators=300	8/12
(ii) mean index	ridge	alpha=0.03	6/12
(iii) first factor	elastic net	alpha=0.03, l1_ratio=0.2	5/12
(iv) k factors	ridge	alpha=0.03	6/12
(v) k factors + covariates	ridge	alpha=1	6/12

G Deviations from the pre-registered design

Supports the reproducibility claims throughout. A study that invokes pre-registration owes the reader an explicit account of where the delivery departed from the plan. Four departures matter. First, the registered primary unit was one row per base model at its best default configuration. The analysis instead uses all 421 configurations. The deduplicated base-model dataset appears below as the registered R1 check, and the configuration level serves as the primary. Second, the registered factor-count rule is parallel analysis, and it retains one factor. The three-factor solution that exhibits economic separation is a pre-declared exploratory over-extraction, and we report H3 as not supported under the registration in consequence. Third, the taxonomy places Terminal-Bench Hard in scientific coding, while the registration grouped it with Terminal-Bench v2.1 as economic. The residualised loadings show Hard at 0.82 on the economic factor, so the assignment is genuinely contestable, and we disclose the split openly. Fourth, the delivered robustness checks depart from the registered suite. We label them distinctly below, and we do not renumber a different battery onto the registered names.

H Robustness checks

Supports Section 5.7. We first report the one registered check that bears directly on the deduplication concern, then the additional checks actually delivered.

Registered R1 (configuration versus deduplicated). We collapse the 421 configurations to one row per base model, keeping the best default configuration as the registration specified. We operationalise it as the highest-intelligence-index row for each base model, after stripping reasoning, effort, and preview suffixes from the model name. This leaves 320 base models, of which 89 carry the complete battery. On this deduplicated grid the first-factor share rises to 90.2% and the date-residualisation drop rises to 24.1 points, and the four economic loadings remain above 0.40. Deduplication therefore strengthens H1 and H2(ii) and leaves the exploratory economic pattern intact, and the configuration-level analysis is the conservative choice for the structural headline. We also re-run the H4 predictive test on this deduplicated grid, where the resampling unit is a distinct base model. The pooled economic-block improvement of the k -factor representation over the single index is +0.038 with a 95% interval of [+0.020, +0.056]. It matches the configuration-level +0.037 and again excludes zero, so the predictive gain does not depend on treating near-duplicate configurations as exchangeable.

Delivered checks C1–C7. (C1) A rank-based factor analysis raises the first-factor share to 94.5%, and (C2) a logit transform of the bounded accuracy benchmarks raises it to 91.8%, so H1 is not a Pearson artefact. (C3) On the 58-model compute-known subsample, the date-only residualisation drops the first-factor share by 16.5 points, and adding log-compute jointly gives a smaller 9.3-point drop, so compute does not deepen the temporal correction beyond date on the same models. (C4) The three clustering algorithms agree on $k = 2$. (C5) The predictor-ladder ordering holds on both the economic block and all twelve targets. (C6) The dominant-factor date fit is robust to the link function. A logistic fit gives $R^2 = 0.505$ against an ordinary-least-squares $R^2 = 0.477$, and both clear 0.30 with the first factor ranked strongest, so H2(i) does not depend on that modelling choice. (C7) The k -factor predictive advantage does not depend on the exploratory factor count. Sweeping the rung-(iv) predictor over $k \in \{2, 3, 4, 5\}$, the improvement over the single index is positive at every count with a bootstrap interval excluding zero. It reads +0.021 at $k = 2$, +0.037 at the $k = 3$ we use, +0.058 at the criterion’s $k = 4$, and +0.059 at $k = 5$, with pooled test R^2 rising from 0.79 to 0.83 across the range. The gain increases with k , so the $k = 3$ we adopt is conservative for this test relative to the criterion’s $k = 4$.

The registered checks not run are R3 (FIML and MICE missingness sensitivity), R4 (replication of the single-factor foil then correction), R5 (reasoning-effort variants as a second scale axis), and R6 (the taxonomy swap test). Each was deferred for space, and each is the natural next step given the deviations above, particularly R5 and R6, which bear on the effort and taxonomy departures.

I Data provenance and reproducibility

Supports Section 2.1 and the reproducibility claims throughout. The primary data set is the Artificial Analysis model leaderboard [20], captured 2026-07-06 as a raw HTML snapshot and pinned by the SHA-256 hash recorded in the shipped manifest (`aa_snapshot_manifest.json`), `6f19f8f08befaaf14cb2d9952e379404c39f246b52d7ae4d1c2aa44795404de6`. The Data API endpoint returned HTTP 401 to our unauthenticated request, so we captured the public leaderboard page and stored the exact bytes. Every downstream number is therefore reproducible regardless of subsequent live-page drift. The secondary data set is Epoch AI Notable AI Models [21], released under CC-BY 4.0 and used only for the scale robustness check. Both snapshots and every result table travel with the submission bundle. The complete pipeline sits in a single notebook, `ME315_240020141.ipynb`, which regenerates all figures and tables from the pinned inputs, and the full code and data will be released at <https://github.com/Looloo-dot/frontier-ai-economic-validity>.

J Supplementary figures

Supports Sections 2.3, 5.1, and 5.4. Figure 6 documents the coverage audit behind the inclusion rule, Figure 5 shows the model clustering, Figure 8 the GDPval regression diagnostics, Figure 9 gives the dimensionality evidence behind H1, and Figure 10 the benchmark-level clustering behind Section 5.4.

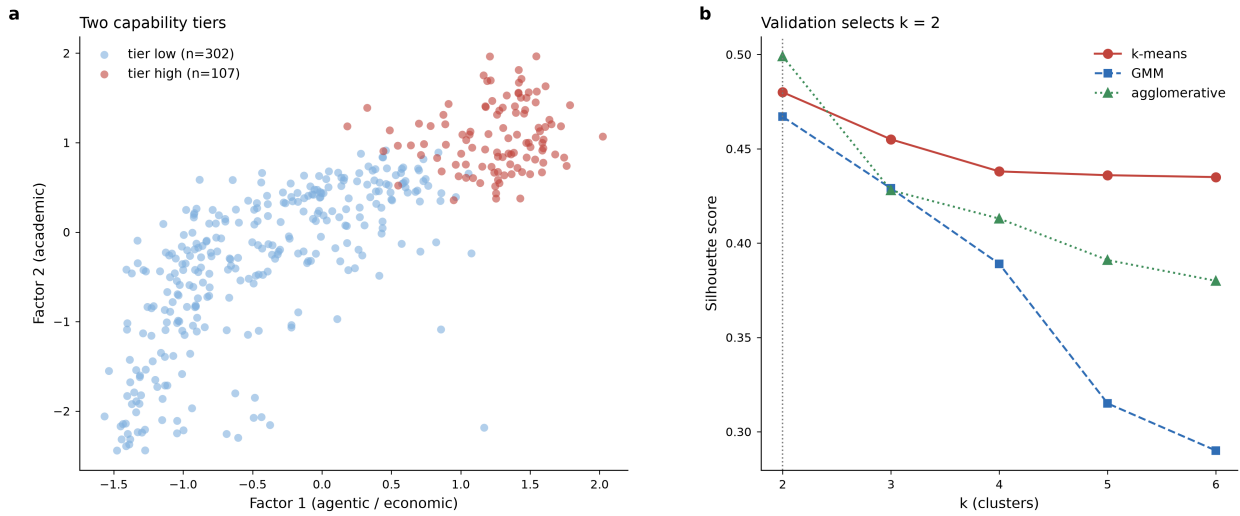


Figure 5: Model clustering on the dense grid ($n = 409$). (a) Models in the space of the first two factors, coloured by the two-cluster agglomerative solution. The factor axes are the dense-grid solution fitted largely without the economic columns, so they track overall capability, and the two clusters mark tiers of it. (b) Silhouette scores by k for all three algorithms agree on $k = 2$.

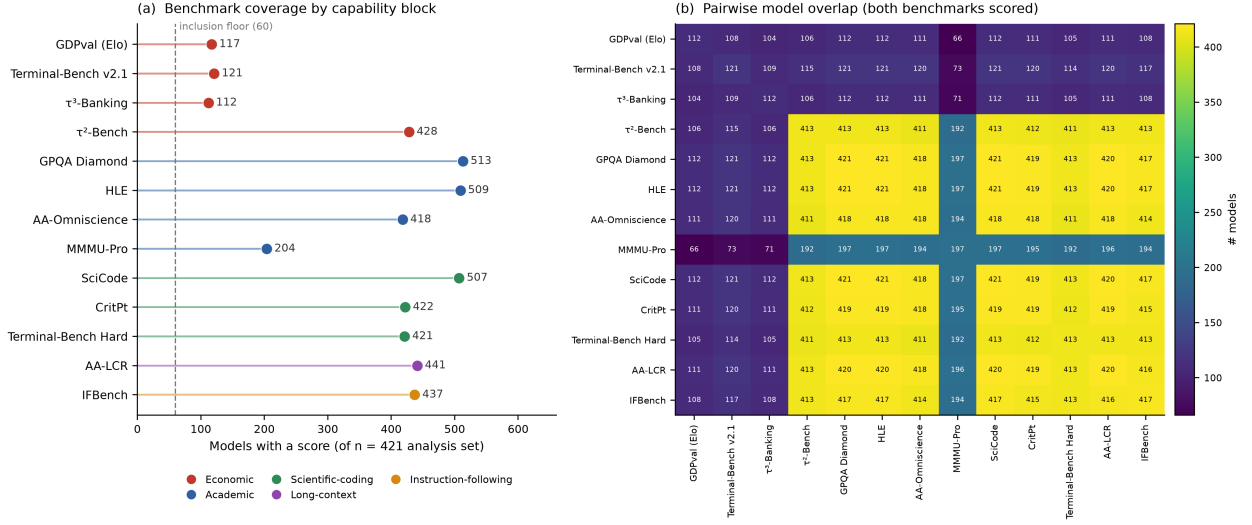


Figure 6: Coverage audit behind the inclusion rule. (a) Number of models carrying a score on each candidate benchmark, coloured by capability block, with the sixty-model inclusion floor marked. Four of the five economic benchmarks are retained. Three of them (Terminal-Bench v2.1, GDPval, and τ^3 -Banking) are scored on only 112 to 121 models, while τ^2 -Bench and the academic core are each scored on four hundred or more. APEX-Agents, with 26 models, falls below the floor and is dropped. (b) Pairwise counts of models scored on both benchmarks in a pair, which set the effective sample size for every correlation in the study. This is the evidence for the two-grid design of Section 2.3.

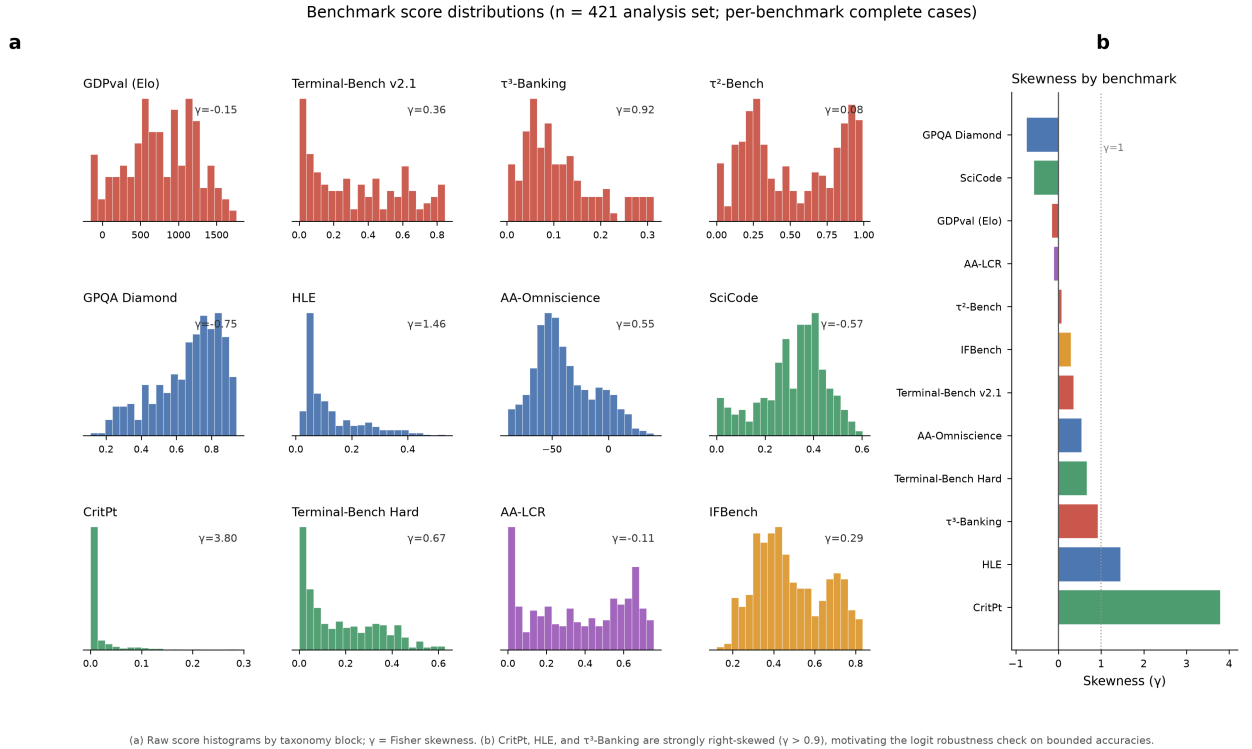


Figure 7: Univariate benchmark score distributions. (a) Raw score histograms for the twelve benchmarks, coloured by capability block; γ is the Fisher skewness. (b) Skewness ordered across benchmarks. Most benchmarks are bounded accuracies in $[0, 1]$, and several are strongly right-skewed (CritPt $\gamma = 3.8$, HLE $\gamma = 1.5$, τ^3 -Banking $\gamma = 0.9$), with scores concentrated near the floor. This boundedness motivates the logit-transform robustness check of Section 5.7. GDPval (Elo) and AA-Omniscience are on unbounded scales and are closer to symmetric.

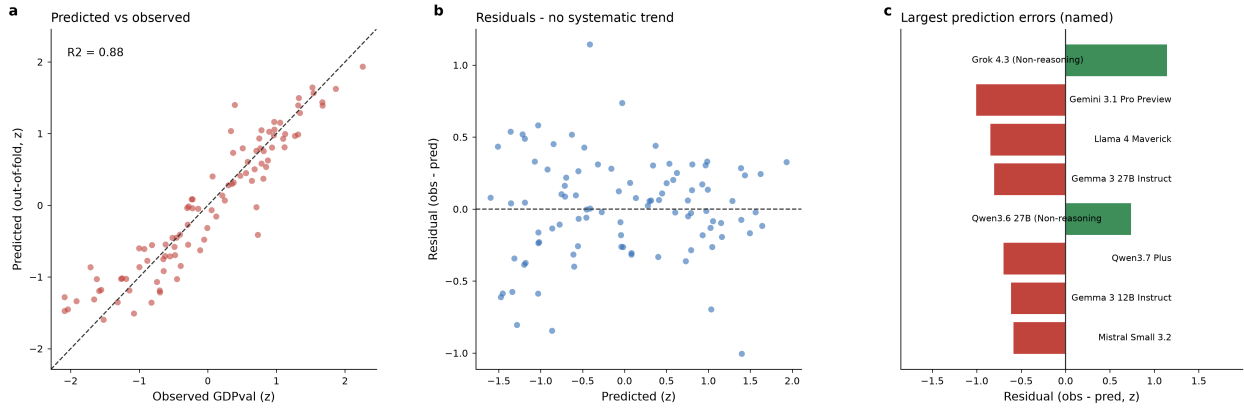


Figure 8: Regression diagnostics for the GDPval k -factor ridge model, out-of-fold. (a) Predicted versus observed, on the identity line with $R^2 = 0.88$. (b) Residuals versus predicted show no systematic trend. (c) The eight largest residuals, named. These are interpretable outliers, and none signals misspecification.

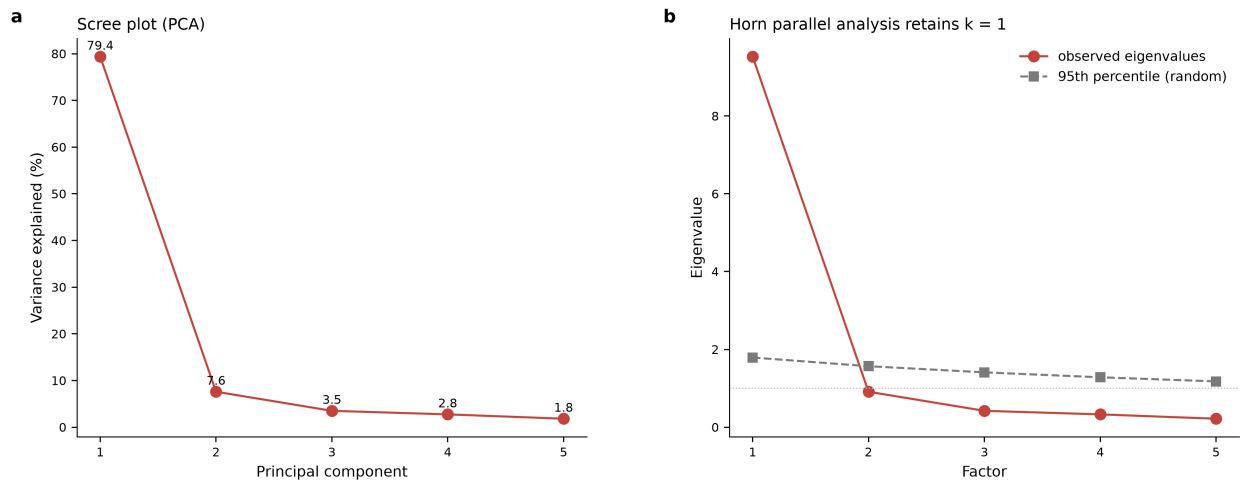


Figure 9: Dimensionality of the battery. (a) The scree plot of principal-component variance shares drops sharply after the first component (79.4%). (b) Horn parallel analysis retains a single factor, as only the first observed eigenvalue (9.53) exceeds its random-data 95th percentile (1.79), while the second (0.91) already falls below (1.57). Values are computed in the notebook and stored in `parallel_analysis.csv`.

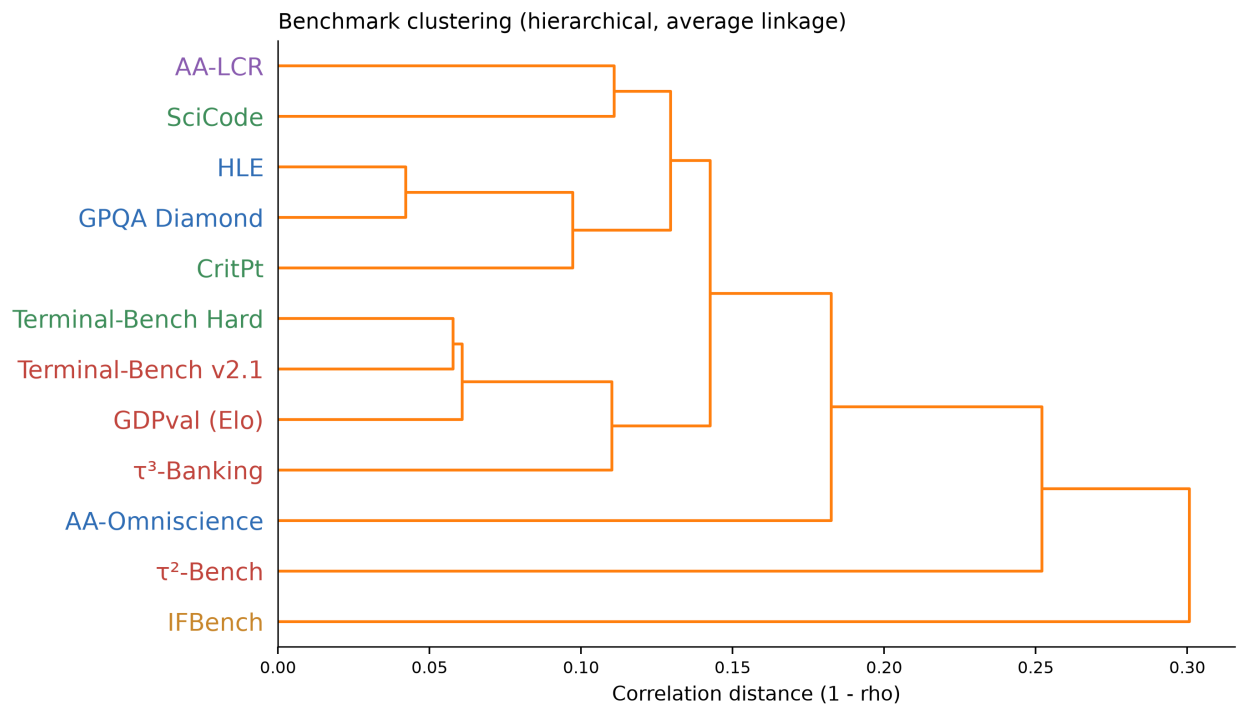


Figure 10: Benchmark clustering by correlation distance ($1 - \rho$, average linkage). Ten of the twelve benchmarks, including three of four economic benchmarks, fall in one block, with τ^2 -Bench and IFBench branching separately.